

List of Tables — Paper / Generation / Experiment Source

All 56 \label{tab:...} entries across jmlr_paper_main, supp_benchmark_report, and supp_routing_improvements, cross-referenced against generate_tables.py. None are currently wired via \input{} — the paper still hard-types every table.

#	Label (tab:)	Caption	Generated file	Status	Experiment / source
1	five_systems_full	Five-System Comparison: Extrap. Error vs. Interp. R ²	five_system.tex	Generated (JSON-backed)	exp1_five_system.py (Core-15); falls back to exp2/exp2_extrap
2	five_systems_performance	Five-System — Performance (Core-15)	five_system_performance.tex	Generated (JSON-backed)	exp1_five_performance.json
3	five_systems_extrapolation	Five-System — Extrapolation (Core-15)	five_system_extrapolation.tex	Generated (JSON-backed)	exp1_five_extrapolation.json
4	five_systems_full_exp2five	Five-System (exp2_five, Feynman suite)	five_system_exp2five.tex	Generated (JSON-backed)	exp2_five own output (five_systems/exp2_five/)
5	five_systems_performance_exp2five	Five-System (exp2_five) — Performance	five_system_exp2five_performance.tex	Generated (JSON-backed)	exp2_five raw protocol_core_noiseless records
6	five_systems_extrapolation_exp2five	Five-System (exp2_five) — Extrapolation	five_system_exp2five_extrapolation.tex	Generated (JSON-backed)	same as row 5
7	main_results	Aggregate extrapolation, HypatiaX DeFi Benchmark (74 tasks)	defi_main.tex	Needs human decision	Paper text carries open [VALUE REDACTED] / reconciliation notes
8	difficulty	Near-perfect success by difficulty, seed 42	defi_tiers.tex	Blocked	Parser always skip_table()s — schema issue unresolved
9	hybrid-bug-breakdown	Hybrid decision-attribution bug breakdown	hybrid_bug_breakdown.tex	Generated (JSON-backed)	hypatiaux_defi_benchmark_v3_results_seed42.json
10	timing_full	Multi-seed timing, both variants	timing_full.tex	Generated (JSON-backed)	10 per-seed hypatiaux_defi_benchmark_{v3c,pca}_seed*.json files
11	timing_llm_routed_full	As above, LLM-routed subset	timing_llm_routed_full.tex	Generated (JSON-backed)	same 10 per-seed JSON files
12	portfolio_seed_sweep	Portfolio Variance seed-sweep	portfolio_sweep.tex	Generated	seed-sweep results
13	llm_ablation	LLM Ablation: PySR vs. HypatiaX, Core-15	ablation.tex	Needs human decision	gen_ablation() uses old mean-row logic; paper table rebuilt with nan/-inf-aware format — out of sync
14	provenance	PCA-split run dates, project history	provenance.tex	Static / stable	transcribed from paper's reviewed text
15	randomsplit	Random 80/20 split, run 2026-07-22	randomsplit.tex	Generated (JSON-backed)	July 22 random-split per-test JSON
16	pcasplit	PCA-directed 40/60 split, run 2026-07-23	pcasplit.tex	Generated (JSON-backed)	July 23 PCA-split per-test JSON
17	feynman30-legacy	Withdrawn 9/30 legacy run (historical only)	feynman30_legacy.tex	Intentionally not automated	source file confirmed absent from released artifacts
18	nguyen12	Nguyen-12 equation table	nguyen12.tex	Needs human decision	never run against real result JSON
19	instability	LLM instability regime distribution	instability.tex	Generated	instability results JSON/CSV
20	methods	Six-method noiseless benchmark	methods.tex	Static / stable	reviewed paper text / config/*.json
21	sweeps	Sweep design	sweeps.tex	Static / stable	reviewed paper text / config/*.json
22	hardcoded	Equations flagged is_hardcoded=True	hardcoded.tex	Generated (JSON-backed)	protocol_core_noiseless_*.json
23	overall	SuppB noiseless overview	suppb_noiseless.tex	Generated	noiseless results data
24	sota	This work's pass rates, 30-eq. benchmark	sota.tex	Generated (JSON-backed)	protocol_core_noiseless_*.json
25	nrmse	Raw vs. normalised RMSE, 30 equations	nrmse.tex	Generated (JSON-backed)	same, rmse/r2 fields
26	wilcoxon	Wilcoxon signed-rank tests	wilcoxon.tex	Generated (JSON-backed)	same, pairwise (stdlib normal approx)
27	r2_noise	R ² statistics per noise level	suppb_r2_noise.tex	Generated	noise-sweep JSON
28	rr_noise	Recovery rate / catastrophic failures per noise level	suppb_rr_noise.tex	Generated	noise-sweep JSON
29	time_noise	Avg. computation time per noise level	suppb_time_noise.tex	Generated	noise-sweep JSON
30	sc_metrics	R ² /RMSE per sample size	suppb_sc_metrics.tex	Generated	sample-complexity JSON
31	winrate	Head-to-head win rates (M3 vs M4)	suppb_winrate.tex	Generated (caveated)	paper calls this “unverified pending exact methodology”
32	arch	Internal routing statistics	arch.tex	Generated (unconfirmed schema)	decision/strategy field tally — schema unconfirmed
33	jmlr	JMLR readiness assessment	jmlr.tex	Static / stable	config/*.json first, else paper text
34	figlist	Complete figure inventory	figlist.tex	Mechanical	lists files actually on disk in figures/
35	software_env	Software/hardware environment	software_env.tex	Static / stable	reviewed paper text

#	Label (tab:)	Caption	Generated file	Status	Experiment / source
36	hyperparameters	PySR configuration	hyperparameters.tex	Static / stable	reviewed paper text
37	hyperparameters_complete	Complete hyperparameter specs	hyperparameters_complete.tex	Static / stable	config/*.json first, else skip_table()
38	llm_detailed	Pure LLM performance by test case	llm_detailed.tex	Generated (JSON-backed)	hypatiax_defi_benchmark_v3*results*.json
39	defi_detailed	DeFi-optimized HypatiaX performance	defi_detailed.tex	Generated (JSON-backed)	same
40	formulas_detailed	Ground truth formulas w/ exact constants	formulas_detailed.tex	Static / stable	config/*.json first, else skip_table()
41	interpolation_stats	Interpolation performance by system	interpolation_stats.tex	Generated (JSON-backed)	protocol_core_noiseless_*.json, r2 field
42	domain_success_detailed	Success rate by domain and method	domain_success_detailed.tex	Generated (JSON-backed)	same, domain-tagged tests
43	validation_stats	Validation-layer error detection stats	validation_stats.tex	Generated (path guessed)	validation_log*.json — no source explicitly named in paper
44	supplementary_files	Supplementary data files	supplementary_files.tex	Mechanical	lists files on disk in output dir
45	additional_stats	Additional statistical tests	additional_stats.tex	Intentionally not automated	no named source anywhere in paper text
46	noise_sensitivity	Success rate vs. noise level	noise_sensitivity.tex	Generated (JSON-backed)	noise_sweep_*.json
47	baseline	Baseline perf. before routing improvements	(none)	Needs human decision	live [VALUE REDACTED] in paper text
48	fix5_cases	Case-level outcomes from Fix 5 / routing guard	fix5_cases.tex	Generated (path guessed)	fix5_cases*.json
49	projected	Measured/projected R ² >0.99 (%) by fix	(none)	Needs human decision	live [VALUE REDACTED] in paper text
50	changes	All changes across two modified files	changes.tex	Intentionally not automated	hand-authored changelog, not a measurement
51	runtime_reproducibility	Software environment (verified 18 Mar 2026)	runtime_reproducibility.tex	Static / stable	reviewed paper text
52	timing_breakdown	Computational cost breakdown	timing_breakdown.tex	Generated (path guessed)	*timing_breakdown*.json
53	scalability	Scalability with dataset size	scalability.tex	Generated (path guessed)	scalability_*.json
54	cost_accuracy_tradeoff	Runtime vs. interpolation tradeoff	(none)	Needs human decision	disputed/removed extrapolation-error figures
55	equation_prevalence	Equation prevalence in training data (estimated)	equation_prevalence.tex	Intentionally not automated	estimate, not a measurement — no source file
56	conceptual_complexity	Conceptual requirements comparison	conceptual_complexity.tex	Intentionally not automated	qualitative comparison — no JSON source

Legend:
generated from real data
static/stable or mechanical
generated but flagged (guessed path / unconfirmed schema / caveated)
blocked or needs a human decision
intentionally never automated